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BUSINESS ANALYSIS ARTEFACT
HEALTHCARE · HEALTHTECH

05

ARTEFACT 05 OF 06

Business Case

The case for a barcode-enabled specimen tracking pilot: problem, options, recommendation and benefits.

PROJECT

Northlake Community Hospital – Specimen Identification & Tracking Improvement

DOCUMENT REF	VERSION	DATE
NCH-BA-05	1.0	October 2026
PREPARED BY	**ROLE**	**PURPOSE**
Michael Solomon	Business Analyst	Investment decision

ⓘ Portfolio case study completed on the 10Alytics Healthcare Business Analysis Program. Northlake Community Hospital is a case-study organisation; all figures are case-study data.

BUSINESS CASE SUMMARY

Barcode-Enabled Specimen Tracking Pilot

Problem statement

Northlake Community Hospital recorded:

52	23	11	2	18
Specimen identification exceptions	Delayed laboratory results	Repeat collections per month	Wrong-patient near misses	Patient complaints

These issues create **patient safety risks** and **increase operational costs**.

Options assessment

Option	Pros	Cons	Cost
1. Process improvement only	Low cost	Risk remains high; manual verification continues	Not costed
2. Barcode tracking pilot Recommended	Increased patient safety; automated verification; improved traceability	—	**CAD 18,500**
3. Full laboratory system replacement	Long-term transformation	High cost; out of scope	Not costed

DECISION

Recommendation

Implement a barcode-enabled specimen verification pilot.

Option 2 addresses the root cause — manual identification and verification — within project scope, at a pilot cost of CAD 18,500.

Benefits

- **Reduced specimen errors**
- **Reduced rework**
- **Faster turnaround times**
- **Stronger audit trail**
- **Improved patient experience**

How success will be measured

KPI	Target
Specimen error rate	< 1%
Turnaround time	≤ 2 hours
Repeat collections	Reduce by 50%
On-time results	≥ 95%
Patient complaints	Reduce by 50%

KPIs and the delivery approach are detailed in Artefact 06 – SDLC Overview.